

HAT-P-18b

R-2445

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-18_150814 · created 2026-08-22T23:34:23+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457248.71571 +/- 0.00066 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.095 +/- 0.052
Transit depth (Rp/Rs)^2	0.91 +/- 0.99 [%]
Orbital Inclination (inc)	86.21 +/- 1.81
Airmass coefficient 1 (a1)	1.8 +/- 1.36
Airmass coefficient 2 (a2)	-0.23 +/- 0.46
Scatter in the residuals of the lightcurve fit is	89.55 %
Best Comparison Star	#3 - [174, 36]
Optimal Aperture	3.36
Optimal Annulus	5.14
Transit Duration (day)	0.042 +/- 0.045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.095 ± 0.052 vs 0.1356 (deviation 0.42σ)
Duration fit vs published	0.042 d vs 0.1129 d (deviation 0.84σ)
Mid-time O–C	-1.1 min
Depth SNR	1.0
Residual scatter	89.55 %

Light curve

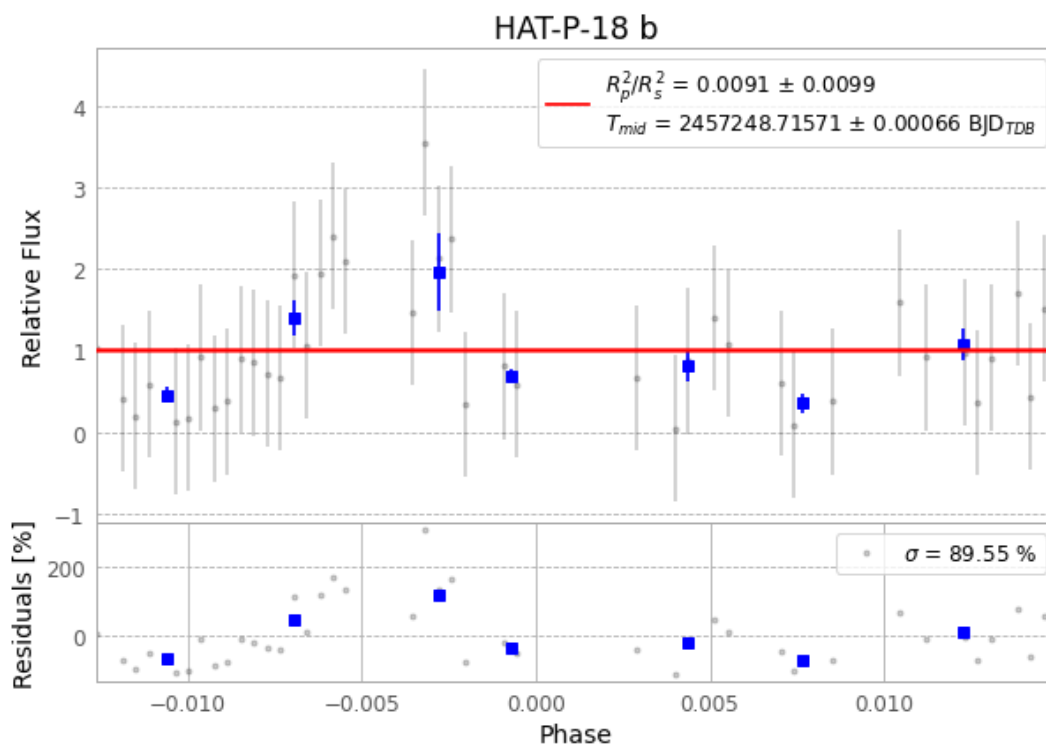


Figure 1 — FinalLightCurve_HAT-P-18 b_2015-08-13.png (SHA-256 c03e772f2bd84afc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [267, 380], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d08336635a420bb7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

78 FITS frames (51 MB), HATP-18150814032113.FITS → HATP-18150814071212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-150814.log (453fe95ec7cb...), FinalLightCurve_HAT-P-18 b_2015-08-13.png (c03e772f2bd8...), FinalLightCurve_HAT-P-18 b_2015-08-13.csv (7db6d345c1bc...)

Compute resources

Wall time 148.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 23:34Z.